

Partitioning of Neutral Organic Compounds to Structural Proteins

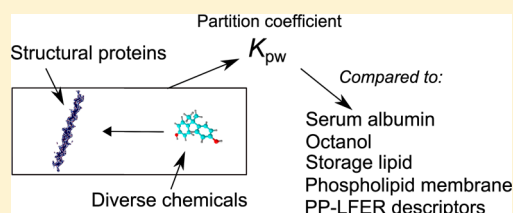
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S Supporting Information

ABSTRACT: Protein–water partition coefficients (K_{pw}) of neutral organic chemicals were measured using muscle proteins (from chicken, fish, and pig), collagen and gelatin. K_{pw} values for these structural proteins were consistently lower than those of bovine serum albumin (BSA), indicating that the use of BSA as a model protein leads to an overestimation of K_{pw} for structural proteins. Differences in K_{pw} between chicken, fish, and pig muscle proteins were small. Across the structural proteins, K_{pw} values were often in the order: muscle proteins > collagen ≥ gelatin. Differences in K_{pw} between the structural proteins were relatively large (<2 log units) for nonpolar compounds, and much smaller or insignificant for polar compounds. There were correlations between $\log K_{pw}$ of muscle proteins and $\log K_{ow}$ ($R_2 = 0.83$ – 0.86 , SD: 0.35 – 0.40 , $n = 45$ – 46). The polyparameter linear free energy relationship (PP-LFER) models fit even better to the data ($R^2 = 0.95$, SD: 0.22). The good model fitting suggests that the reversible binding to muscle proteins can be considered to be nonspecific binding. There was an indication that some chemicals may sorb irreversibly to muscle proteins, which needs further research. We found that the partitioning to muscle protein is typically weaker than that to lipids, but that the protein partitioning of an H-bond donor compound can be as strong as the storage lipid partitioning.



INTRODUCTION

The extent of the partition of chemicals into organisms and their tissues has direct implications in bioaccumulation, toxicokinetics, and toxic effects of the chemicals. In environmental studies, it is often assumed that the sorptive capacity of a given organism or tissue for neutral organic chemicals can fully be represented by its lipid fraction.^{1–3} The other biological constituents including proteins are assumed to have negligible contributions to the total sorptive capacity of the organism/tissue. Some researchers, however, argued that proteins in animals could also be a significant accumulation phase for organic compounds, particularly in lean tissues such as muscle.^{4–6} A thorough investigation on the partitioning of chemicals to tissue proteins has not yet been conducted though.

Equilibrium partitioning of a chemical to a protein is described by the protein–water partition coefficient K_{pw} [$L_{\text{water}}/\text{kg}_{\text{protein}}$], defined as follows:

$$K_{pw} = C_{\text{protein}}/C_{\text{water}} \quad (1)$$

where C_{protein} and C_{water} are the chemical's equilibrium concentrations in protein and water, respectively, at a specified temperature. Experimental data for K_{pw} are limited in comparison to the partition coefficients to lipids, for which a large body of data are available.^{7,8} An exception is serum albumin: this blood plasma protein has been extensively investigated with regard to its sorptive properties (e.g., refs 9 and 10) because of its large influence on the bound/unbound fractions in plasma and thus on the pharmaceutical and toxic activities of the chemicals. In addition to its own relevance,

serum albumin is often considered as a first model protein to investigate protein affinity of chemicals.¹¹ However, K_{pw} may well depend on the types of proteins, as the amino acid composition and the three-dimensional structure can differ considerably across different proteins. Serum albumin is a water-soluble globular protein having known ligand-binding sites,^{12,13} whereas structural proteins, the most abundant type of proteins in animal body, are typically present as bundled fibers or filaments. Thus, the binding properties of structural proteins can be substantially different from those of serum albumin.

K_{pw} data for structural proteins are very limited. Some values for collagen are available in the literature.^{14–17} However, these values may be inaccurate due to the influence of residual lipids in the collagen samples (see the Results and Discussion Section and the Supporting Information (SI) for details). Recently, deBruyn and Gobas⁵ attempted to obtain K_{pw} values for animal tissue protein by analyzing literature-reported tissue-blood distribution coefficients. deBruyn and Gobas,⁵ in principle, subtracted the contributions of lipids and water from the total sorptive capacities of tissue and assigned the remaining sorptive capacity to proteins. However, their approach⁵ has the following shortcomings: (i) It is assumed that blood and tissue proteins have the identical sorptive capacity. (ii) All lipids including storage and membrane lipids are modeled with the

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octanol–water partition coefficient, K_{ow} (see ref 7 for more details). (iii) Lipids predominate the tissue sorptive capacity in most of the cases they analyzed and, thus, subtracting the lipid contribution often does not leave any significant sorptive capacity that could be assigned to proteins. In general, a value that results from the difference of two close values suffers from a large relative uncertainty. This problem explains why they falsely reported a number of negative values for the protein/lipid sorptive capacity ratio ($Z_{\text{protein}}/Z_{\text{lipid}}$), which is equivalent to the lipid–protein partition coefficient and cannot be negative by definition. In the end, deBruyn and Gobas⁵ recommended to estimate K_{pw} as 5% of the lipid–water partition coefficient, or octanol–water partition coefficients (K_{ow}), for modeling purposes, but this relationship needs re-evaluation. More recently, Hansen et al.¹⁸ assembled a large set of K_{pw} data for keratin, the main protein component of, for example, skin, hair, nails, wool, and feathers. The data collected in ref 18 were, however, measured under very different conditions using different original materials and consist mainly of complex pharmaceutical compounds including those that are ionic under the experimental conditions.

The first purpose of this study was to determine K_{pw} of neutral organic chemicals using structural proteins in a consistent manner. Muscle protein, collagen, and gelatin were used as working materials. Muscle protein is made up 60–95% of actin and myosin,¹⁹ which form filaments serving as the main constituents of muscle fiber. Muscle protein is particularly relevant for the assessment of bioaccumulation, because muscle is a major part eaten by predators (including humans). In addition, muscle is a lean tissue where the relative contribution of proteins to the overall sorptive capacity of the tissue may be comparably high. In contrast to muscle proteins, collagen is principally a single type of protein bearing minor variations. Collagen is ubiquitously found in all animal tissue, is the most abundant vertebrate protein, and is the major component of connective tissues such as tendon, cartilage, and bone.¹⁹ A collagen molecule has a triple-helix structure consisting of three left-handed helical polypeptides. Collagen triple helices bundle up and form a strong fibrous structure. Gelatin is produced by partial hydrolysis of collagen, where covalent and hydrogen bonds are broken by thermal and chemical treatments and the triple helix structure is destabilized.²⁰ Gelatin does not occur in nature, but we used it because gelatin dissolves in water, whereas collagen and muscle proteins principally do not, and the change in binding properties from collagen to gelatin may be mechanistically interesting. Following the experimental work, the K_{pw} values for various proteins including serum albumin were compared to examine the protein-to-protein variability. Subsequently, estimation models for K_{pw} were presented. Finally, K_{pw} values for muscle protein were evaluated in comparison to partition coefficients to lipids.

MATERIALS AND METHODS

Protein Samples. Chicken, fish, and pig muscle proteins were prepared from chicken breast meat, Pacific cod fillet, and pork fillet, respectively, purchased from a local grocery store. Visible fat tissues were removed using tweezers. Chicken and pork were homogenized using an Ultra Turrax homogenizer (IKA, Staufen, Germany) under cooling in an ice–water bath. The fish sample was homogenized with a meat grinder because the Ultra Turrax homogenization was hampered by an extensive foam formation. Homogenized samples were freeze-dried, ground with a mortar to remove large aggregates, and

delipidated by solvent extraction as described below and in the SI. A collagen sample originating from bovine Achilles tendon, which contains mainly collagen Type I, was obtained from Sigma (C9879, lot 017K7018) and was also delipidated by solvent extraction. Gelatin powder was obtained from Merck (Darmstadt, Germany; Emprove exp Ph Eur, BP, NF grade) and was used as purchased. Table S1 in the SI summarizes the sample information.

Delipidation of Samples. Samples were delipidated by repeated extraction with a 2:1 chloroform/methanol (C/M) mixture. The C/M mixture has been shown to be suitable for extracting both nonpolar and polar lipids from meat.^{21,22} For each sample, 5–10 g were extracted with at least 3 times ≥ 200 mL C/M. More information on the extraction procedure is given in the SI. The extracts containing lipids were filtered and concentrated by a nitrogen stream and vacuum until no weight change occurred in order to determine the lipid contents in the original samples. The extracted protein samples were dried under a nitrogen stream at least for half a day, vacuumed overnight or longer, ground with a mortar, and stored at -30 °C until use.

Determination of Residual Lipid Content. Residual lipid contents in the delipidated samples were measured to ensure the removal of lipid. Since lipid contents in defatted samples are extremely low, the conventional gravimetric method could not be used. Therefore, the lipid content was determined by digesting the lipids and measuring the released fatty acids using a gas chromatograph equipped with a flame ionization detector (see the SI for the experimental procedure). Performance of the method was evaluated by measuring the lipid contents of pure olive oil and a palmitic acid-spiked fish muscle protein, which confirmed a recovery of $93 \pm 3\%$ ($n = 2$) and $94 \pm 2\%$ (mean $\pm$ SD, $n = 3$), respectively. This method can detect a fatty acid content of as low as 0.01% (w/w) in the sample. Disadvantages of the method are: the method does not distinguish different types of lipids (e.g., triglycerides, phospholipids, free fatty acids), lipids that do not contain fatty acids in the molecular structure are neglected, and nonfatty acid parts of lipid molecules (e.g., phosphate) are neglected, too. Nevertheless, this method is considered satisfactory for indicating very low lipid contents in extracted protein samples.

Batch Partitioning Experiments. Batch partitioning experiments were conducted to determine K_{pw} values. Batch experiments for chicken and fish muscle proteins and collagen were performed using 47–64 diverse neutral organic compounds to address the influences of molecular structure on K_{pw} . Additionally, K_{pw} values for pig muscle protein and gelatin were measured using 10–14 compounds for comparison to the above-mentioned proteins. The experimental procedure has been described before for our work with serum albumin.¹⁰ Briefly, a protein sample was weighed into a glass vial or test tube, which received buffer solution containing 150 mM NaCl, 10 mM phosphate (pH 7.40), and 200 mg L⁻¹ NaN₃. Vials with gelatin were heated at 50 °C for 10 min to dissolve gelatin completely in buffer. Amounts of protein in the suspension were 0.004 to 0.1 g_{protein}/mL_{buffer}. Gelatin solutions did not show gelation during the experiment. A mixed methanol stock solution of several test compounds was added to the protein suspensions and solutions. Solute loadings on proteins at equilibrium were 10–400 mg/kg_{protein}. Batches were horizontally shaken (160 rpm) at 37 °C. Preliminary experiments were conducted to examine the equilibrium times (see SI). Volatile compounds were measured via the headspace after an

equilibration time of 1 day. The other compounds were measured by an equilibrium passive sampling method using a polymer-coated microfiber with an equilibration time of 2 days (except for PAHs). GC/MS (7890A/5975C, Agilent Technologies, Inc., Wilmington, DE) was used for quantification. GC/MS conditions were described elsewhere.²³ K_{pw} was obtained from the mass balance calculation based on the mass conservation assumption, that is, the concentration decrease in the water phase occurs solely due to the partitioning to protein. Air–water and fiber–water partition coefficients were used to take into account the mass fractions in the respective phases.¹⁰ Data were not considered further if the protein-sorbed fraction was <20% of the mass of the test chemical added to the system.

Recovery Test. As described above, K_{pw} was determined based on the mass conservation assumption. Most of the chemicals used here were stable in buffer and serum albumin solution, as confirmed in our previous work.^{10,23} Thus, a significant mass loss due to volatilization or sorption to a third phase (e.g., septum) is unlikely. However, if transformation of chemicals occurs due to the presence of the structural protein sample, the K_{pw} value determined from the mass balance calculation is higher than the true value. Moreover, it is of interest to know whether the binding to proteins is a reversible process. Therefore, recovery tests were performed with chicken and fish muscle proteins and collagen following the method described in ref 10. Here only briefly: spiked protein suspensions were liquid–liquid-extracted with isohexane, cyclohexane, or ethyl acetate. Vials were shaken only gently to prevent foam formation. Because of the gentle shaking a prolonged extraction time (>3 days) was given for extraction. The extracts were analyzed with GC/MS. Ethyl acetate extracts were dried with sodium sulfate anhydrous before GC analysis. Recovery was calculated as,

$$\text{recovery} = M_{\text{suspension}} / (M_{\text{total}} - M_{\text{fiber}}) \quad (2)$$

where $M_{\text{suspension}}$, M_{total} , and M_{fiber} are, respectively, the solute mass in the protein suspension measured by solvent extraction, the solute mass added to the batch, and the solute mass measured in the fiber (which is 0 for the headspace method).

All recovery data are shown in SI Table S2. In the following discussions, all K_{pw} data accompanied with recovery >90% are considered, and K_{pw} data with 80–90% recovery are included only if the sorbed fraction is >40%. Data with recovery <80% were not considered. Many compounds had to be removed from consideration because of the low recovery or unsuccessful measurements of recovered extracts. Note that selecting data this way implies that only reversible binding to proteins is considered. Note also that no recovery experiment was done for nonpolar or weakly polar compounds (e.g., alkanes, alkyl and halogenated benzenes), as these compounds are unlikely to transform or irreversibly bound to proteins in our experimental system. Recovery of PAHs was measured, and the three large PAHs showed relatively low recovery (SI Table S2). This is likely because of very slow extraction of these compounds by mild shaking with organic solvent and is not caused by degradation, as has been observed for recovery from natural organic matter suspensions.²⁴ Thus, the K_{pw} data for all measured PAHs are used for further analysis. Recovery data for alcohols are largely unavailable, because the GC peak shapes were substantially poor due probably to residual water or matrix in the extract. Nevertheless, the data for alcohols are included because K_{pw} data appear to be consistent, as suggested by the

linear increase of $\log K_{pw}$ with the increasing carbon number. Finally, recovery tests were not performed for pig muscle protein and gelatin. For them, no transformation and irreversible binding of a given solute was assumed if high recovery was obtained for the other proteins.

RESULTS AND DISCUSSION

Influence of Residual Lipids on the Partition Coefficients. Impurity lipids in protein samples could hamper the experimental determination of K_{pw} , because partitioning to proteins can be weaker than that to lipids, and thus a relatively small amount of residual lipids can have a significant influence on the measured partition coefficients. Preliminary experiments for 10 compounds showed that partition coefficients to the original, nonextracted collagen sample are 1–2 log units higher than the C/M-extracted collagen (SI Figure S3), coinciding with a relatively high lipid content (i.e., 17%(w/w)) of the original collagen sample. Similarly, an isohexane-extracted collagen exhibited around 1 log unit higher partition coefficients than the C/M-extracted collagen, which is consistent with the fact that apolar solvents do not extract polar lipids completely.^{21,22} These results highlight the importance of thoroughly removing lipids from protein samples for investigating the organic solute partitioning to structural proteins. Residual lipid analysis showed that our C/M-extracted samples contain only ≤0.1% lipids (SI Table S1) and are thus suitable for K_{pw} determination.

A few K_{pw} data for collagen are available in the literature.^{14–17} However, the reported values are up to ca. 2 log units higher than the values determined in this study using the C/M-extracted collagen, as detailed in the SI. Because all these studies^{14–17} used collagen samples from Sigma without any lipid removal, influences of residual lipid may explain the high K_{pw} values reported in these studies.

Comparison between Different Types of Proteins. The measured K_{pw} values that reflect reversible binding to chicken and fish muscle proteins are shown in Table 1. All other measured values are presented in SI Tables S3 and S4. Since the sorption to collagen was weaker than expected, we succeeded to obtain K_{pw} values only for 11 compounds (SI Table S3).

In the literature, serum albumin is often used to investigate the general protein affinity of chemicals, assuming that K_{pw} is not strongly dependent on the type of protein. Experimental partition coefficients to bovine serum albumin (BSA) (from our previous work)¹⁰ and to chicken muscle protein are both available for 40 compounds with varying polarity. The comparison reveals that K_{pw} values for BSA are generally higher than for chicken muscle protein (Figure 1). The difference is on average 0.83 log units. The largest difference is 1.8 log units, found for 1,4-dibromobenzene. The higher K_{pw} values for BSA can be explained by the specific binding sites of BSA, as such sites are absent in muscle proteins. The amino acid compositions of BSA and muscle protein are fairly similar (see the SI) and thus cannot be a main reason for the differences in K_{pw} . For two compounds (cyclooctane, chrysene), K_{pw} for BSA and chicken muscle protein are similar, within 0.1 log units. We hypothesize that this similarity is due to the reduced sorption to BSA of these compounds because of their unfavorable molecular shape to fit to the binding pocket(s) of BSA (also see ref 10 for more discussions).

Figure 2 compares K_{pw} of six selected compounds for BSA and all structural proteins studied here (see SI Figure S4 for all

Table 1. Log K_{pw} for Chicken and Fish Muscle Proteins^a

	chicken muscle protein	fish muscle protein
	log $K_{pw} \pm SD$ (n)	log $K_{pw} \pm SD$ (n)
n-nonane	3.86 ± 0.11 (4)	3.68 ± 0.09 (4)
cyclooctane	2.94 ± 0.02 (4)	2.82 ± 0.05 (4)
cycloheptane	2.16 ± 0.01 (4)	2.33 ± 0.03 (4)
1-nonene	3.54 ± 0.07 (4)	3.52 ± 0.07 (4)
hept-1-yne	1.75 ± 0.04 (4)	1.62 ± 0.02 (4)
1-chlorooctane	2.93 ± 0.01 (4)	2.88 ± 0.05 (4)
di-n-butyl ether	1.72 ± 0.01 (4)	1.52 ± 0.08 (4)
2-octanone	1.57 ± 0.03 (4)	1.35 ± 0.06 (3)
2-nonanone	1.83 ± 0.02 (4)	1.51 ± 0.06 (4)
2-decanone	2.11 ± 0.06 (4)	1.96 ± 0.06 (4)
1-nitrohexane	1.34 ± 0.05 (4)	
tri-n-butyl phosphate	1.69 ± 0.03 (4)	1.83 ± 0.01 (3)
1-hexanol		0.55 ± 0.05 (5)
1-octanol	1.42 ± 0.02 (3)	1.55 ± 0.03 (5)
1-nonanol	1.97 ± 0.04 (4)	1.99 ± 0.05 (5)
4-ethyl-3-hexanol		0.75 ± 0.02 (5)
n-propylbenzene	1.98 ± 0.03 (4)	1.95 ± 0.04 (4)
n-pentylbenzene	2.71 ± 0.06 (4)	2.77 ± 0.02 (5)
n-hexylbenzene	3.19 ± 0.05 (4)	3.24 ± 0.02 (5)
1,2,4-trimethylbenzene	1.90 ± 0.02 (4)	1.87 ± 0.02 (4)
1,2-dichlorobenzene	2.02 ± 0.02 (4)	2.04 ± 0.03 (4)
1,2,3,4-tetrachlorobenzene	2.80 ± 0.01 (4)	2.86 ± 0.03 (5)
1,2,4,5-tetrachlorobenzene	2.94 ± 0.02 (4)	3.00 ± 0.03 (5)
pentachlorobenzene	3.35 ± 0.04 (4)	3.41 ± 0.03 (5)
1,4-dibromobenzene	2.18 ± 0.02 (4)	2.28 ± 0.03 (4)
dibenzothiophene	3.40 ± 0.02 (15)	3.46 ± 0.01 (5)
phenanthrene	3.14 ± 0.03 (15)	3.25 ± 0.01 (5)
pyrene	3.67 ± 0.03 (15)	3.76 ± 0.02 (5)
chrysene	4.53 ± 0.05 (10)	4.41 ± 0.04 (5)
benzo[a]pyrene	4.88 ± 0.05 (10)	4.97 ± 0.03 (5)
4-nitroanisole	1.38 ± 0.07 (4)	
valerophenone	1.74 ± 0.03 (4)	1.75 ± 0.01 (6)
benzophenone	2.01 ± 0.03 (4)	1.99 ± 0.01 (6)
ethyl benzoate		1.27 ± 0.02 (3)
di-n-propyl phthalate		1.92 ± 0.01 (6)
2-nitrotoluene	1.27 ± 0.03 (4)	1.32 ± 0.02 (3)
2,4-dinitrotoluene	1.22 ± 0.04 (4)	1.27 ± 0.01 (3)
1-nitronaphthalene		2.16 ± 0.01 (3)
4-n-propylphenol	1.69 ± 0.01 (4)	1.63 ± 0.02 (5)
4-fluorophenol	1.18 ± 0.00 (3)	1.10 ± 0.01 (5)
3-chlorophenol	1.57 ± 0.02 (4)	1.55 ± 0.02 (5)
4-chlorophenol	1.55 ± 0.02 (3)	1.45 ± 0.02 (5)
4-bromophenol	1.70 ± 0.02 (4)	1.62 ± 0.01 (5)
4-iodophenol	1.93 ± 0.02 (3)	1.84 ± 0.02 (4)
2-phenylphenol	2.00 ± 0.03 (4)	
4-chloroaniline	1.19 ± 0.03 (3)	
4-iodoaniline	1.66 ± 0.05 (3)	
4-nitroaniline	0.84 ± 0.01 (3)	
N,N-diethylaniline	1.60 ± 0.02 (3)	1.52 ± 0.01 (5)
indole	1.40 ± 0.01 (4)	1.97 ± 0.01 (5)
diazepam	2.01 ± 0.01 (4)	2.10 ± 0.03 (4)

^aOnly those that reflect reversible binding are presented. See Table S3 for an extended data list.

16 compounds for which this comparison is possible). For all compounds, K_{pw} of BSA is the highest of the proteins considered, indicating that the use of BSA as a model protein generally leads to an overestimation of K_{pw} for structural proteins.

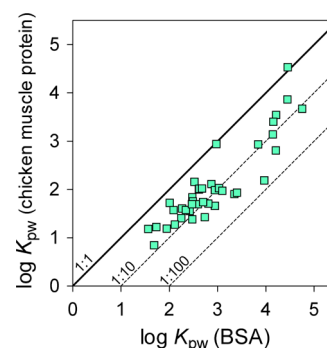


Figure 1. Comparison of K_{pw} values measured with chicken muscle protein and bovine serum albumin (BSA). Data for BSA are from Endo and Goss.¹⁰

Differences in K_{pw} between chicken, fish, and pig muscle proteins are marginal for the compounds studied (Figure 2, SI Figure S4). An excellent 1:1 agreement of K_{pw} values for chicken and fish muscle proteins (average absolute error: 0.10 log units; $n = 40$; SI Figure S5) also shows the high similarity of these proteins. These results suggest that K_{pw} for muscle proteins generally is independent of the species. Though confirmed for only three quite distinct species now, such a relationship would be extremely useful, as it allows us to extrapolate the K_{pw} values to other species.

Focusing on the structural proteins investigated here: K_{pw} values are often in the order, muscle proteins > collagen ≥ gelatin (Figure 2, SI Figure S4). Differences in K_{pw} between these structural proteins are distinct for nonpolar compounds such as alkyl and halogenated benzenes (up to 2 log units). For polar compounds, variations in K_{pw} appear to be smaller (e.g., valerophenone, indole). Particularly, for H-bond donor polar compounds (e.g., phenols), the partitioning to gelatin is somewhat stronger than that of collagen and is comparable to that of muscle proteins. The relatively high K_{pw} values observed for muscle proteins cannot readily be explained, as muscle protein is a mixture of various types of proteins. The observed differences between collagen and gelatin can be at least qualitatively explained as following: gelatin is a hydrolysis product of collagen, thus is more polar functionalized and more hydrated in water, which diminishes the interactions with nonpolar compounds in comparison to collagen. In contrast, for polar compounds this hydration effect is counteracted with enhanced polar interactions, leading to similar K_{pw} of gelatin and collagen.

Model Evaluation: Correlation with Log K_{ow} . A simple linear regression using log K_{ow} as the descriptor was fitted to the log K_{pw} data for chicken and fish muscle proteins (Figure 3). Log K_{pw} values are from Table 1 and log K_{ow} values are experimental values from the KOWWIN database²⁵ amended with four estimated values.²⁶

$$\begin{aligned} \log K_{pw}(\text{chicken muscle protein}) \\ = 0.73(\pm 0.04)\log K_{ow} - 0.39(\pm 0.17) \\ (n = 46, R^2 = 0.86, \text{SD} = 0.35) \end{aligned} \quad (3)$$

$$\begin{aligned} \log K_{pw}(\text{fish muscle protein}) \\ = 0.80(\pm 0.06)\log K_{ow} - 0.68(\pm 0.21) \\ (n = 45, R^2 = 0.83, \text{SD} = 0.40) \end{aligned} \quad (4)$$

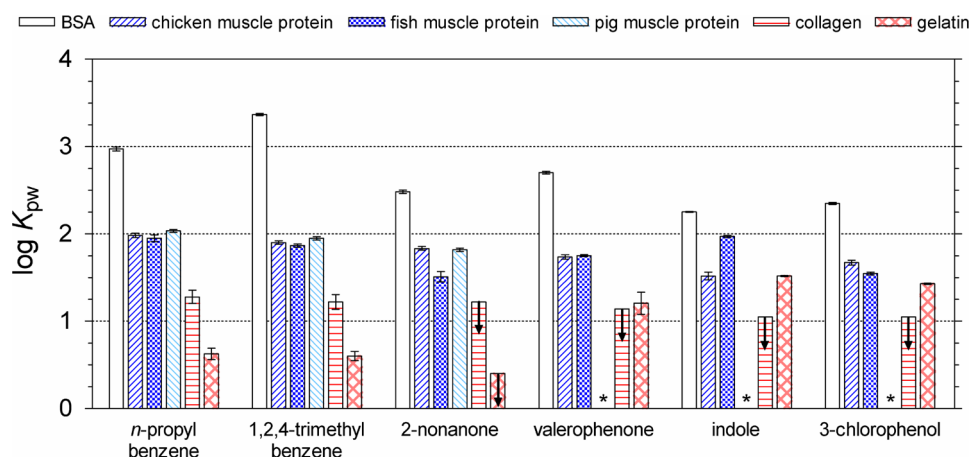


Figure 2. Measured K_{pw} of selected, representative compounds for different proteins. Error bars indicate standard deviations. Stars indicate no data. Arrows indicate that K_{pw} values were too small to measure and that the actual K_{pw} values should be lower than indicated by the bars. Figures for additional compounds are presented in the SI.

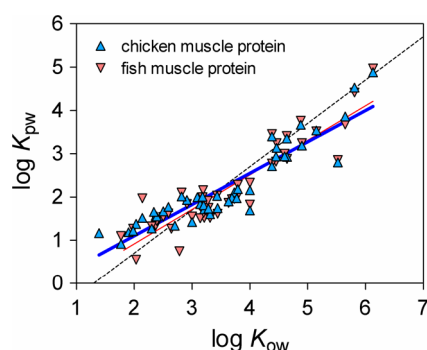


Figure 3. Correlations between $\log K_{pw}$ for muscle proteins and $\log K_{ow}$. The thick blue line is the linear regression for chicken muscle protein, and the thin red line the linear regression for fish muscle protein. The dashed line is the relationship from deBruyn and Gobas⁵ (i.e., $K_{pw} = 0.05 K_{ow}$).

The slight differences in the values of fitting parameters are not significant, as seen in the overlapping of the two regression lines in Figure 3. It appears that the relationship proposed by deBruyn and Gobas⁵ (i.e., $K_{pw} = 0.05 K_{ow}$) is not an unreasonable approximation for muscle proteins (Figure 3), although the derivation of this relationship is disputable (see the Introduction section). Note that eq 3 (or eq 4) and the " $K_{pw} = 0.05 K_{ow}$ " relationship would lead to larger differences if extrapolated to more hydrophobic compounds ($\log K_{ow} > 6$). Neither of the models is however validated for such compounds. The regressions from Hansen et al.¹⁸ for keratin (slope: 0.27–0.34; intercept: + 0.64–1.24) are fairly different from eqs 3 and 4. Such difference can occur for various reasons such as the different experimental setting and materials used to obtain experimental data. Types of calibration compounds used could also contribute to bias in the regression coefficients, as the correlation between $\log K_{pw}$ and $\log K_{ow}$ is not perfect.

$\log K_{pw}$ – $\log K_{ow}$ correlations for pig muscle protein, collagen, and gelatin are presented in SI Figure S6. R^2 was 0.82, 0.43, and 0.31, respectively, showing comparatively weak correlations for collagen and gelatin. However, the number of data points for these three samples is limited (i.e., 9–12 compounds) and thus, the presented relationships could be biased.

Polyparameter Linear Free Energy Relationships (PP-LFERs). As an alternative model, a PP-LFER equation was fitted to the K_{pw} data for chicken and fish muscle proteins. The PP-LFER model used is as following,

$$\log K_{pw} = c + eE + sS + aA + bB + vV \quad (5)$$

The dependent variables are E , excess molar refraction; S , dipolarity/polarizability parameter; A , solute H-bond acidity; B , solute H-bond basicity; and V , molar volume. The values of the descriptors used are presented elsewhere.^{10,23} For details about the model, see publications from Abraham such as refs 27 and 28. Multiple linear regression analysis for $\log K_{pw}$ of chicken and fish muscle proteins resulted in the following equations,

$$\begin{aligned} \log K_{pw}(\text{chicken muscle protein}) = & -0.79(0.25) \\ & + 0.51(0.10)E - 0.51(0.17)S + 0.26(0.17)A \\ & - 2.98(0.24)B + 3.01(0.21)V \quad (n = 46, \\ & R^2 = 0.95, \text{SD: } 0.22) \end{aligned} \quad (6)$$

$$\begin{aligned} \log K_{pw}(\text{fish muscle protein}) = & -0.99(0.24) \\ & + 0.54(0.11)E - 0.46(0.18)S + 0.30(0.18)A \\ & - 3.19(0.23)B + 3.12(0.20)V \quad (n = 45, \\ & R^2 = 0.95, \text{SD: } 0.22) \end{aligned} \quad (7)$$

Values in parentheses are standard errors. Octanal was removed from the data set for fish muscle protein, because it was a clear outlier with a fitting error of 0.9 log unit. Values of the fitting coefficients were similar between chicken and fish, as expected from the similar experimental data. Good fitting was indicated in Figure 4. An SD of 0.22 is typical for fitting of PP-LFER models to homogeneous phases like solvents and rubbery polymers. The fitting quality is an indication that the binding mechanisms to muscle proteins are nonspecific and not influenced by additional interaction mechanisms that are not accounted for by PP-LFER models. The fitting of the PP-LFER model to muscle protein- K_{pw} is substantially better than that of BSA- K_{pw} .¹⁰ This difference agrees with the line of argument that serum albumin binding is specific to some extent.¹⁰ Note that such PP-LFER-based analysis provides a more solid indication of specific binding of proteins than a

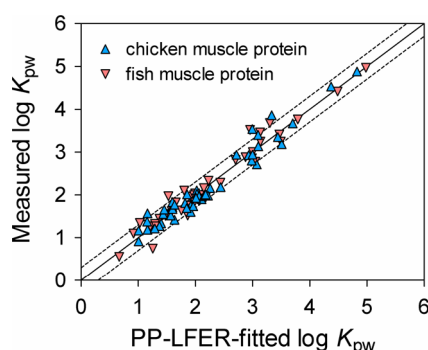


Figure 4. Fitting of PP-LFERs models to $\log K_{pw}$ for muscle proteins. The solid line indicates the 1:1 agreement, and the dashed lines 0.3 log unit deviations.

mere comparison of individual K_{pw} values could do, because the magnitude of K_{pw} can differ for many reasons (e.g., different extent of nonspecific polar and nonpolar interactions). Another type of PP-LFER²⁹ that uses L , the log of the hexadecane–air partition coefficient, instead of E is presented in the SI together with the results of the fitting to K_{pw} .

It should be stressed that the K_{pw} data used for LFER model calibration reflect only reversible binding. Any irreversible binding mechanism (e.g., covalent bond formation) would lead to higher K_{pw} values than estimated from the LFER models established above. Candidate compounds that may undergo such a process are those compounds for which the recovery test exhibited low recovery. In SI Figure S7, mass-balance-based experimental (apparent) K_{pw} values of low recovered-compounds (i.e., < 80% recovery) are compared with PP-LFER-predicted K_{pw} . For chicken muscle protein, the experimental and PP-LFER predicted K_{pw} are generally in good agreement, except for the two aldehydes, for which experimental values are 0.8 or 1.0 log unit higher than the PP-LFER predictions. For fish muscle protein, experimental values of many low recovered-compounds are substantially higher than the PP-LFER predicted values; differences are 1–1.2 log units for the two aldehydes, the two nitroalkanes, and 1-naphthol, and 0.5–1 log unit for 4-nitroaniline, 4-chloroaniline, 4-iodoaniline, 4-aminobiphenyl, 2,4,6-trinitrotoluene, metolachlor, and atrazine. Some of the chemicals named above might undergo irreversible binding with muscle proteins. Propachlor is another candidate, for which the fiber phase concentration was lower than the quantification limit (i.e., apparent K_{pw} was too high to measure). Note again that the experimental approach of this study cannot differentiate between degradation and irreversible binding, thus the discrepancy discussed above could partially be due to degradation of chemicals during the experiment. Further work to differentiate these two processes as well as to explore which types of chemicals can form irreversible bonds with proteins can be justified.

Comparison with Lipids. To evaluate the sorptive capacity of muscle proteins in comparison to lipids, K_{pw} for chicken muscle protein was compared to storage lipid–water partition coefficients ($K_{\text{storage lipid/water}}$) and phospholipid membrane–water partition coefficients ($K_{\text{membrane/water}}$). The PP-LFER models for chicken muscle protein (eq 6), storage lipid,⁸ and phospholipid membrane⁷ were used to derive the respective partition coefficients. The comparison was made for 633 selected compounds including a wide variety of size and polarity. Figure 5 shows the results for the $\log K_{pw}$ range of 0–6, which reveals: (i) For nonpolar and H-bond acceptor polar

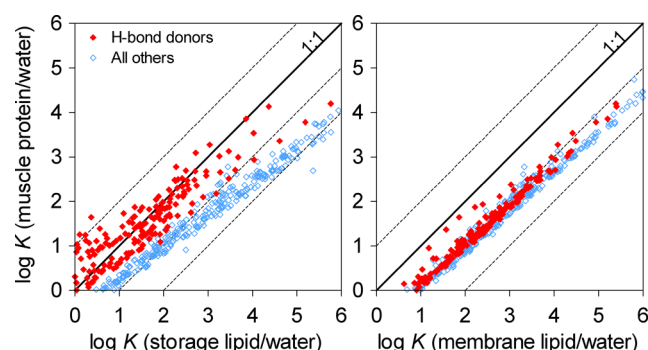


Figure 5. Partition coefficients of a variety of compounds to chicken muscle protein and to storage and membrane lipids. The partition coefficients are calculated using the respective PP-LFER equations. H-bond donor polar compounds ($A > 0.2$) are marked in red.

compounds, K_{pw} of muscle protein is 1–2 log units smaller than $K_{\text{storage lipid/water}}$ and the difference grows with increasing magnitude of the partition coefficients, (ii) H-bond donor compounds (e.g., phenols, anilines, many polar pesticides and pharmaceuticals) exhibit comparable values of K_{pw} and $K_{\text{storage lipid/water}}$ (mostly within ± 1 log unit), (iii) K_{pw} for muscle protein is consistently smaller than $K_{\text{membrane/water}}$ by 1–2 log units for any type of compounds. Proteins contain a number of peptide bonds ($-\text{CO}-\text{NH}-$) and various polar side groups in their molecular structure, which can all act as H-bond acceptor sites. Thus, it is reasonable that the $K_{pw}-K_{\text{storage lipid/water}}$ relationship of H-bond donor compounds differs from that of the other compounds. The strong correlation between $\log K_{pw}$ and $\log K_{\text{membrane/water}}$ which is even stronger than the correlation between $\log K_{pw}$ and $\log K_{ow}$ is an interesting coincidence because phospholipid membrane and muscle protein are structurally unrelated.

Based on the relatively high affinity of H-bond donor compounds for proteins and the low lipid content of the original fish muscle, it is anticipated that the partitioning of H-bond donor compounds to the original, lipid-containing fish muscle is predominated by its protein fraction. To test this expectation, partition coefficients from water to freeze-dried, nonextracted fish muscle ($K_{\text{dry fish muscle/water}}$) were measured for eight H-bond donor compounds. Indeed, $K_{\text{dry fish muscle/water}}$ values generally agree with K_{pw} within 0.2 log units (SI Figure S8). Indole and 1-nonanol showed $K_{\text{dry fish muscle/water}}$ values being 0.4–0.5 log units higher than K_{pw} , probably because they are relatively weak H-bond donors and some contributions from lipids exist. The good agreement between $K_{\text{dry fish muscle/water}}$ and K_{pw} for fish muscle also indicates that the solvent-extraction does not have a major influence on the binding property of muscle proteins.

This study offers the first comprehensive investigation on the sorption capacity of structural proteins. The data clearly show that serum albumin is not an optimal model protein to represent affinity of chemicals for structural proteins. Muscle protein may be suggested as a starting material to investigate sorptive capacities of structural proteins, because muscle protein is easy to prepare, relevant for food-web bioaccumulation, and abundant in animals. The observed good correlations with $\log K_{ow}$ and PP-LFER descriptors indicate that the reversible binding to muscle protein is nonspecific. Formation of irreversible bonding was suggested for some compounds, which warrants further study. Comparison with lipids suggests that partitioning to structural proteins can be as

high as that to storage lipid for H-bond donor compounds. Of course, the relative contribution of proteins to the overall partitioning capacity of an organism or a tissue depends also on the relative compositions of proteins, lipids, and water and this issue will be addressed in an upcoming article.

■ ASSOCIATED CONTENT

● Supporting Information

Further information on the experimental methods and comparison with literature data; four additional tables and eight additional figures. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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